In [1]:
```
import seaborn as sns
import matplotlib.pyplot as plt
```

In [2]:
```
df = sns.load_dataset('titanic')
```

In [3]:
```
df.head()
```

Out[3]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_m
0	0	3	male	22.0	1	0	7.2500	S	Third	man	T
1	1	1	female	38.0	1	0	71.2833	C	First	woman	Fa
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	Fa
3	1	1	female	35.0	1	0	53.1000	S	First	woman	Fa
4	0	3	male	35.0	0	0	8.0500	S	Third	man	T

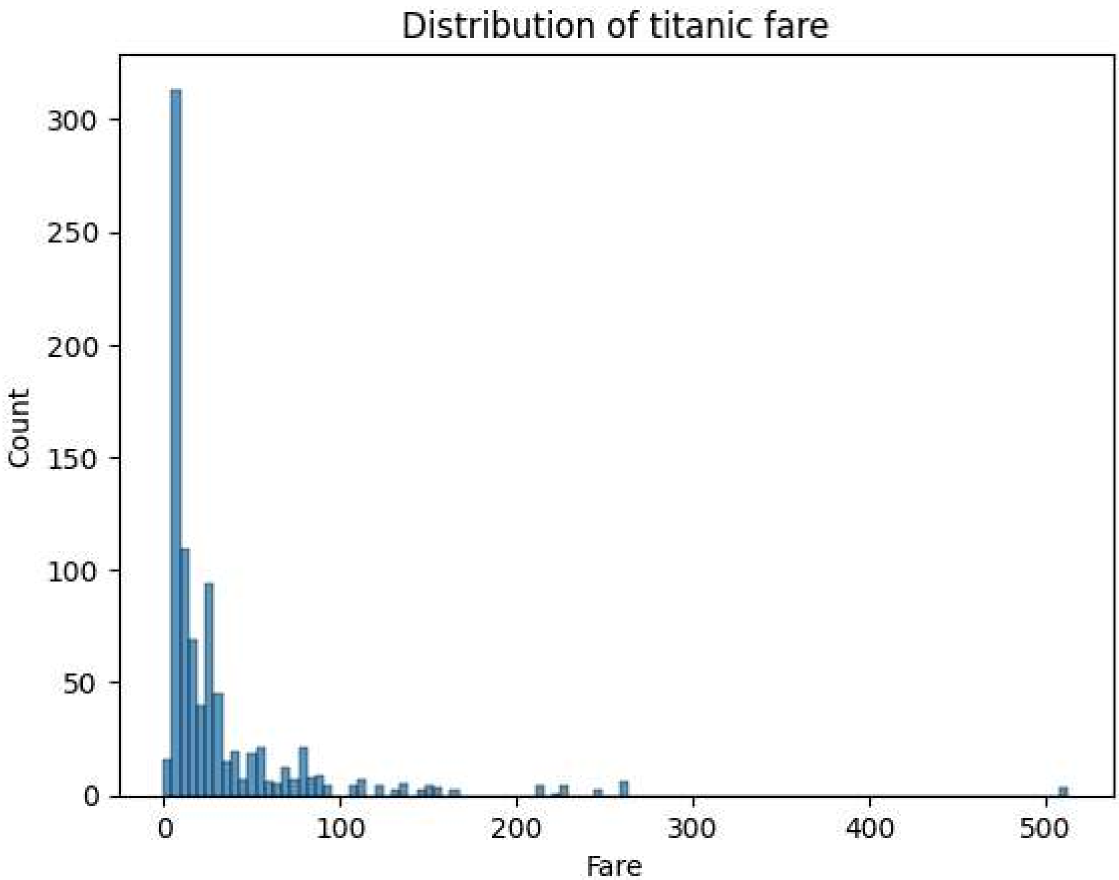
In [5]:
```
df.describe
```

Out[5]:
```
<bound method NDFrame.describe of      survived  pclass     sex   age  sibsp  parc
h     fare embarked   class  \
0           0       3    male  22.0      1      0   7.2500        S   Third
1           1       1  female  38.0      1      0  71.2833        C   First
2           1       3  female  26.0      0      0   7.9250        S   Third
3           1       1  female  35.0      1      0  53.1000        S   First
4           0       3    male  35.0      0      0   8.0500        S   Third
..        ...     ...     ...   ...    ...    ...      ...      ...     ...
886         0       2    male  27.0      0      0  13.0000        S  Second
887         1       1  female  19.0      0      0  30.0000        S   First
888         0       3  female   NaN      1      2  23.4500        S   Third
889         1       1    male  26.0      0      0  30.0000        C   First
890         0       3    male  32.0      0      0   7.7500        Q   Third

       who  adult_male deck  embark_town alive  alone
0      man        True  NaN  Southampton    no  False
1    woman       False    C    Cherbourg   yes  False
2    woman       False  NaN  Southampton   yes   True
3    woman       False    C  Southampton   yes  False
4      man        True  NaN  Southampton    no   True
..     ...         ...  ...          ...   ...    ...
886    man        True  NaN  Southampton    no   True
887  woman       False    B  Southampton   yes   True
888  woman       False  NaN  Southampton    no  False
889    man        True    C    Cherbourg   yes   True
890    man        True  NaN   Queenstown    no   True

[891 rows x 15 columns]>
```

In [6]:
```
df.shape
```

Out[6]: (891, 15)

In [8]:
```
sns.histplot(df['fare'])
plt.title("Distribution of titanic fare")
plt.xlabel("Fare")
plt.ylabel("Count")
plt.show()
```



In the Titanic fare histogram, patterns mean:

trends or behaviors visible in the data distribution

When you plot:

```
sns.histplot(df['fare'])
```

you can visually observe important characteristics of the dataset.

Main Patterns You Can Find

---

1. Most Passengers Paid Low Fare

You will see:

- Tall bars on left side
- Near smaller fare values

Meaning:

majority passengers bought cheaper tickets

Interpretation

This suggests:

- Most passengers belonged to lower classes
- Cheap tickets were more common

---

2. Very Few Passengers Paid High Fare

On right side:

- Small bars appear far away

Meaning:

only few passengers paid expensive fares

These may belong to:

```
First-class passengers
```

---

3. Positive (Right) Skewness

Histogram usually shows:

- Long tail on right side

Meaning:

fare distribution is positively skewed

Why Positive Skew Happens?

Because:

- Most fares are small
- Few fares are extremely high

So data stretches towards higher values.

---

4. Presence of Outliers

Very high fare values appear isolated.

These are:

outliers

Example:

```
Fare > 500
```

may appear very far from most data.

---

5. Uneven Distribution

Data is not equally spread.

This tells:

Titanic fare does not follow normal distribution

What External Expects When Asking “Find Patterns”

You should mention:

- Distribution shape
- Concentration of data
- Skewness
- Outliers
- Frequency trends

Perfect Viva Explanation

You can say:

> “From the histogram, we can observe that most passengers paid lower fares, while only a few passengers paid very high fares. The distribution is positively skewed because the histogram has a long right tail. Some extreme fare values can also be identified as outliers.”

How Histogram Helps Find Patterns

Histogram helps identify:

- Frequency concentration
- Data spread
- Skewness
- Outliers
- Shape of distribution

Additional Pattern You Can Mention

If bars are:

- Very concentrated → low variability
- Widely spread → high variability

Common Viva Questions

Q. What pattern is visible in Titanic fare?

Most passengers paid low fare and distribution is positively skewed.

Q. Why histogram useful?

It helps understand distribution and frequency patterns.

Q. What indicates skewness in histogram?

Long tail on one side.

Q. How to identify outliers in histogram?

Extreme isolated bars far from majority data.

Q. Why are most bars near left side?

Because most passengers paid lower fares.

Advanced Viva Point

You can also say:

> "The histogram suggests socioeconomic inequality among passengers because only a small number could afford high ticket fares."

This sounds very impressive in viva.

DSBDA Practical 8 — Data Visualization I

Titanic Fare Distribution using Seaborn Histogram

Dataset Used:

`Titanic Dataset`
We will use:

`Seaborn inbuilt Titanic dataset`

Aim

To analyze Titanic passenger fare distribution using:

- Histogram
- Seaborn library

What is Data Visualization?

Data Visualization means:

- Representing data graphically
- Understanding patterns and trends visually

Examples:

- Histogram
- Bar chart
- Scatter plot
- Pie chart

What is Seaborn?

Seaborn is a Python visualization library built on:

`Matplotlib`
Used for:

- Statistical graphics
- Attractive plots
- Easy visualization

About Titanic Dataset

Contains passenger details:

- Age
- Sex
- Fare
- Survival status
- Passenger class

Rows:

```
891
```

What is Fare?

`fare`
represents:

ticket price paid by passenger

What is Histogram?

Histogram is used to:

- Show frequency distribution
- Analyze continuous numerical data

Histogram Theory

- X-axis → Data values
- Y-axis → Frequency/count

It groups data into intervals called:

```
bins
```

Step 1: Import Libraries

Code

```
import seaborn as sns
import matplotlib.pyplot as plt
```

Explanation

Library	Purpose
seaborn	Visualization
matplotlib	Plot display

Step 2: Load Titanic Dataset

Code

```
df = sns.load_dataset('titanic')
```

Explanation

`load_dataset()`

Loads Seaborn inbuilt datasets.

Display Dataset

Code

```
df.head()
```

Step 3: Check Dataset Information

Code

```
df.info()
```

Explanation

Displays:

- Columns
- Datatypes
- Missing values

Step 4: Plot Histogram

Code

```
sns.histplot(df['fare'])

plt.title("Distribution of Titanic Fare")

plt.xlabel("Fare")
plt.ylabel("Count")

plt.show()
```

Explanation of Each Command

`sns.histplot()`

Creates histogram.

`df['fare']`
selects fare column.

`plt.title()`

Adds graph title.

`plt.xlabel()`

Labels X-axis.

`plt.ylabel()`

Labels Y-axis.

`plt.show()`

Displays graph.

Expected Observation

You will observe:

- Most passengers paid lower fares
- Few passengers paid very high fares

This means:

Fare distribution is positively skewed

What is Positive Skewness?

When:

- Most values concentrated on left
- Tail extends towards right

Why Titanic Fare is Skewed?

Because:

- Many passengers traveled in lower class
- Few passengers bought expensive tickets

Improved Histogram with KDE

Code

```
sns.histplot(df['fare'], kde=True)

plt.title("Fare Distribution with KDE")

plt.show()
```

What is KDE?

KDE = Kernel Density Estimation

It shows:

- Smooth probability density curve

Changing Number of Bins

Code

```
sns.histplot(df['fare'], bins=30)

plt.show()
```

What are Bins?

Bins are intervals/groups used in histogram.

More bins:

- Detailed distribution

Fewer bins:

- Simpler distribution

Important Theory

Difference Between Histogram and Bar Chart

Histogram	Bar Chart
Continuous data	Categorical data
Bars touch each other	Bars separated

What Does Histogram Show?

- Distribution
- Shape
- Spread
- Skewness
- Outliers

What is Distribution?

Arrangement/spread of data values.

Types of Distribution

Type	Shape
Normal	Bell-shaped
Positively skewed	Right tail
Negatively skewed	Left tail

Important Viva Questions

Basic Questions

What is histogram?

Graph showing frequency distribution.

Why histogram used?

To analyze distribution of numerical data.

What does fare represent?

Ticket price.

Which library used?

Seaborn and Matplotlib.

What is seaborn?

Python statistical visualization library.

Intermediate Questions

What are bins?

Intervals in histogram.

Why bars touch in histogram?

Because data is continuous.

What is KDE?

Smooth density curve.

What type of distribution does fare have?

Positively skewed distribution.

Advanced Questions

Why fare distribution is skewed?

Few passengers paid very high fares.

Difference between histogram and distplot?

histplot	distplot
New function	Deprecated

Why visualization important?

Helps identify patterns and trends.

External May Ask Modifications

Plot Histogram for Age

```
sns.histplot(df['age'])

plt.show()
```

Add Color

```
sns.histplot(df['fare'], color='red')

plt.show()
```

Add KDE and Bins Together

```
sns.histplot(df['fare'], bins=20, kde=True)

plt.show()
```

Plot Using Matplotlib

```
plt.hist(df['fare'])

plt.show()
```

Difference Between Seaborn and Matplotlib

Seaborn	Matplotlib
High-level	Low-level
Better styling	More control

Common Exam Errors

Error	Reason
Dataset not loading	Internet issue
Wrong column name	Case-sensitive
plt.show missing	Plot not displayed

2-Minute Viva Explanation

“In this practical, we used Seaborn’s inbuilt Titanic dataset and visualized the distribution of passenger ticket fares using a histogram. First, we imported Seaborn and Matplotlib libraries and loaded the dataset using load_dataset(). Then we plotted the histogram of the fare column using histplot(). The histogram showed that most passengers paid lower fares while very few paid higher fares, indicating a positively skewed distribution. Histograms are useful for understanding frequency distribution and spread of continuous numerical data.”